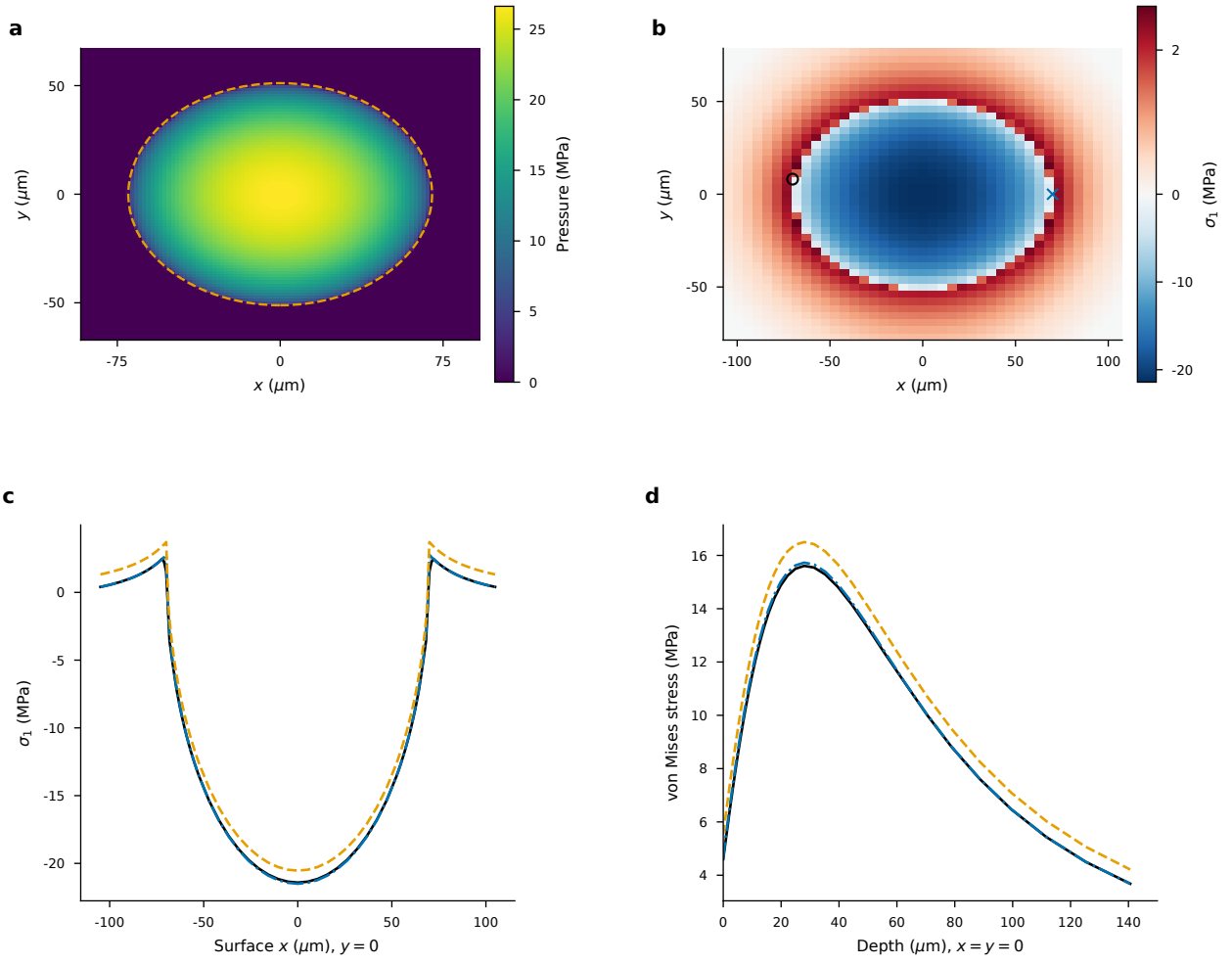

a
50
0
-50
y (μm)
-75
0
75
x (μm)
25
20
15
10
5
0
Pressure (MPa)
b
50
0
-50
y (μm)
-100
-50
0
50
100
x (μm)
2
0
-10
-20
σ1 (MPa)
c
0
-5
-10
-15
-20
σ1 (MPa)
-100
-50
0
50
100
Surface x (μm), y = 0
d
16
14
12
10
8
6
4
von Mises stress (MPa)
0
20
40
60
80
100
120
140
Depth (μm), x = y = 0
Reference
Hertz + shell residual
Hertz half-space
Observed reference variation